Math Practice Test B

Decimals, percentages and word problems · No calculator

Show your working. Write your final answer, with units, in the answer box.

1 Work out 15.1 – 7.46

Working

Answer

2 Work out 0.6 × 0.07

Working

Answer

3 Work out 4.75 × 8

Working

Answer

4 Work out 6.08 + 12.7 + 3.425

Working

Answer

5 Work out 13.5 ÷ 6

Working

Answer

6 36 students went on a field trip. This was 45% of the grade. How many students are in the grade?

Working

Answer

7 A bike costs $240. The price goes up by 10%. Then the new price is reduced by 25%. What is the price now?

Working

Answer

8 Movie tickets cost $8.75 for an adult and $5.40 for a child. A family buys 2 adult and 3 child tickets and pays with $40. How much change do they get?

Working

Answer

9 Sara is 6 years older than Tom. Their ages add up to 30. How old is Sara?

Working

Answer

10 A pen costs 3 times as much as a pencil. Together they cost $2.40. How much does the pen cost?

Working

Answer

Answer Key: Math Practice Test B

Answers involving money or ages should include units ($, ¢, years). Working shows one way to solve each.

Q	Answer	Working	Skill checked
1	**7.64**	15.10 – 7.46 (add a zero first)	Subtract multi-digit decimals (6.NS.B.3)
2	**0.042**	6 × 7 = 42, three decimal places, so 0.042 (0.42 is wrong)	Multiply multi-digit decimals (6.NS.B.3)
3	**38**	475 × 8 = 3800, two decimal places, so 38.00 = 38	Multiply multi-digit decimals (6.NS.B.3)
4	**22.205**	Line up the points: 6.080 + 12.700 + 3.425	Add multi-digit decimals (6.NS.B.3)
5	**2.25**	13.50 ÷ 6 = 2.25 (add a zero to keep dividing)	Divide a decimal by a whole number (6.NS.B.3)
6	**80 students**	5% = 36 ÷ 9 = 4, so 100% = 80	Find the whole given a part and the percent (6.RP.A.3.c)
7	**$198.00**	$240 + 10% = $264; $264 – 25% = $198 (not –15% overall, which gives $204)	Solve multi-step percent problems (7.RP.A.3)
8	**$6.30**	2 × $8.75 = $17.50; 3 × $5.40 = $16.20; total $33.70; $40 – $33.70 = $6.30	Multi-step money problem with decimals (7.NS.A.3)
9	**18 years**	Tom = t, Sara = t + 6. 2t + 6 = 30, so t = 12, so Sara is 18	Write and solve an equation from a word problem (7.EE.B.4)
10	**$1.80**	Pencil = p, pen = 3p. 4p = 240¢, so p = 60¢, pen = $1.80	Write and solve an equation from a word problem (7.EE.B.4)